**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 282 · Network types, architecture and hardware

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does an OS device driver do?

Q2. What does a router connect?

Q3. Define a client-server relationship.

Q4. Binary search on sorted indexes 0–14 for 13: show midpoint index sequence.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

LAN — Local Area Network.

WAN — Wide Area Network.

switch — Forwards frames inside local network.

router — Forwards packets between IP networks.

access point — Provides wireless connection to LAN.

## New learning

Understand — A LAN covers a local area such as a school building; a WAN connects networks across wider geography.

Trace — Client-server uses dedicated service systems; peer-to-peer allows peers to exchange roles and resources, with different management implications.

Explain — NICs connect devices, switches forward local Ethernet frames, wireless APs bridge Wi-Fi, and routers forward IP packets between networks.

## Worked example

Trace school laptop reaching an external site through a wired school network.

1. Predict — Switch moves frames within LAN

2. Check the answer — router handles packets destined for another network.

# We Do · Practise with guidance

Two classrooms use Wi-Fi but need access to a central school file server. Identify useful hardware/architecture.

```
Clients → access points → switch → file server.
```

Predict

APs provide wireless access

Check the answer

switch connects local devices

client-server centralises authentication/files.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Define LAN, WAN, switch and router. [4]

### Question 2

Describe a laptop's network path via wireless access point, switch and router to the Internet. [6]

### Question 3

Compare client-server and peer-to-peer for school file sharing, giving strengths and limits. [6]

## Exit Ticket · Answer before checking

A network switch and router are always interchangeable because they both move packets.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Offers an interface to hardware.

Q2. Different IP networks.

Q3. Client requests service from a dedicated server.

Q4. mid7, mid11, mid13; target found at index13.

### You Do pages · Independent questions

Q1. LAN local network [1], WAN wide-area connection [1], switch forwards LAN frames [1], router joins IP networks [1].

Q2. NIC/Wi-Fi interface [1], AP receives frame [1], switch forwards on LAN [1], router selects external network [1], ISP/WAN carries traffic [1], return follows appropriate route [1].

Q3. Client-server central storage/permission policies [2] but server dependency/cost [1]; P2P direct resource sharing [2] but decentralised management [1].

### Exit Ticket · Model answer

False. Their usual layer and forwarding responsibilities differ.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick correct answers; rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding